

JavaScript Drill: Part II

Lab2.1 Variables and Data Types

Objective

Declare and display variables of different data types.

Instructions

1. Declare variables: name (string), age (number), isStudent (boolean), grades (array), student (object).
2. Print them and their types using `console.log` and `typeof`.

Code:

```
let name = "Juan";
let age = 20;
let isStudent = true;
let grades = [85, 90, 92];
let student = {firstName: "Juan", lastName: "Dela Cruz"};

console.log(name, typeof name);
console.log(age, typeof age);
console.log(isStudent, typeof isStudent);
console.log(grades, typeof grades);
console.log(student, typeof student);
```

Expected Output

- Console shows variable values and their corresponding types.

Lab 2.3: Arithmetic & Type Conversion

Objective

Perform arithmetic operations and demonstrate type conversion.

Instructions

1. Ask the user for two numbers using `prompt()`.
2. Perform addition, subtraction, multiplication, and division.
3. Display results in an alert.
4. Show conversion of a string to a number and a number to boolean.

Code:

```
let num1 = Number(prompt("Enter first number:"));
let num2 = Number(prompt("Enter second number:"));

let sum = num1 + num2;
let diff = num1 - num2;
let prod = num1 * num2;
let div = num1 / num2;

alert(`Results:
Sum = ${sum}
Difference = ${diff}
Product = ${prod}`);
```

```
Division = ${div}`);

let numString = "123";
let numValue = Number(numString);
let boolValue = Boolean(0);

console.log("String:", numString, "Type:", typeof numString);
console.log("Converted Number:", numValue, "Type:", typeof numValue);
console.log("Boolean Value of 0:", boolValue);
```

Expected Output

- Alert box shows arithmetic results.
- Console logs conversion results.

Lab 2.4: Conditional Statements

Objective

Use comparison and logical operators in conditionals.

Instructions

1. Ask the user for their age.
2. If 18 or older, display “You are an adult.” Otherwise, display “You are a minor.”
3. Validate the input to ensure it’s numeric.

Code:

```
let ageInput = prompt("Enter your age:");
let ageNum = Number(ageInput);

if (!isNaN(ageNum) && ageNum >= 18) {
    alert("You are an adult.");
} else if (!isNaN(ageNum)) {
    alert("You are a minor.");
} else {
    alert("Invalid input!");
}
```

Expected Output

- User is classified as adult, minor, or given Invalid input! message.

Lab 2.5: Loops

Objective

Practice for and while loops.

Instructions

1. Use a for loop to display numbers 1–10 in the console.
2. Ask the user for a number and compute its factorial using a while loop.

Code:

```
// For loop
for (let i = 1; i <= 10; i++) {
    console.log(i);
}

// Factorial using while loop
let n = Number(prompt("Enter a number to compute factorial:"));
let fact = 1;
let i = 1;
while (i <= n) {
    fact *= i;
    i++;
}
console.log(`Factorial of ${n} = ${fact}`);
```

Expected Output

- Console logs numbers 1–10.
- Console shows factorial of the input number.

Lab 2.6: Arrays and Objects

Objective

Manipulate arrays and objects.

Instructions

1. Create an array of fruits with three elements and add another using `.push()`.
2. Remove the first fruit using `.shift()`.
3. Create an object `student` with properties `name`, `age`, and `course`.
4. Print both the array and object to the console.

Code:

```
let fruits = ["apple", "banana", "mango"];
fruits.push("orange");
fruits.shift();
console.log(fruits);

let student = {
    name: "Maria",
    age: 21,
    course: "IT"
};
console.log(student);
```

Expected Output

- Console shows updated fruits array.
- Console shows student object.